



Founded 1982

STUDENT NAME: \_\_\_\_\_

TEACHER: \_\_\_\_\_

## THE HILLS GRAMMAR SCHOOL

### TASK 4 Trial Examination 2015

### YEAR 12

# MATHEMATICS

**Time Allowed:** Three hours (plus five minutes reading time)

**Weighting:** 40%

**Instructions:**

- Approved calculators may be used
- Attempt all questions
- Start all questions on a new sheet of paper
- The marks for each question are indicated on the examination
- Show all necessary working

| MCQ | Question 11 | Question 12 | Question 13 | Question 14 | Question 15 | Question 16 | TOTAL |
|-----|-------------|-------------|-------------|-------------|-------------|-------------|-------|
| 10  | 15          | 15          | 15          | 15          | 15          | 15          | 100   |



**Section I****10 marks****Attempt Questions 1-10****Allow about 15 minutes for this section**

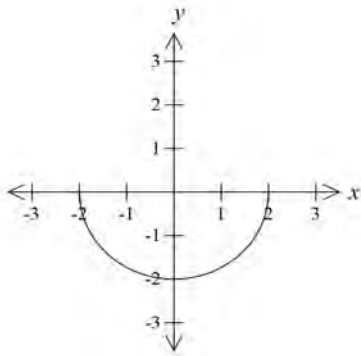
Use the multiple-choice answer sheet for Questions 1-10.

- 1 What is the value of  $\frac{(1.49)^2 - 1.98}{\sqrt{11.62 + 8.34 \times 2.72}}$  correct to three significant figures?

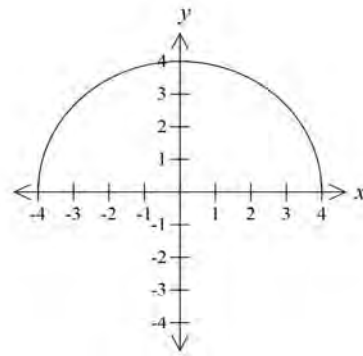
- (A) 0.040  
 (B) 0.0410  
 (C) 0.0409  
 (D) 0.041

- 2 Which graph best represents  $y = \sqrt{4 - x^2}$  ?

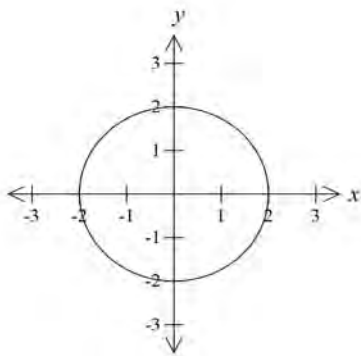
(A)



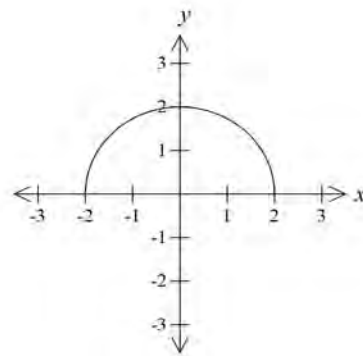
(B)



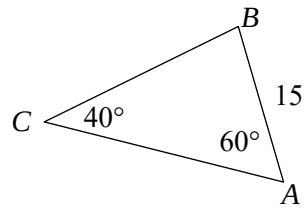
(C)



(D)

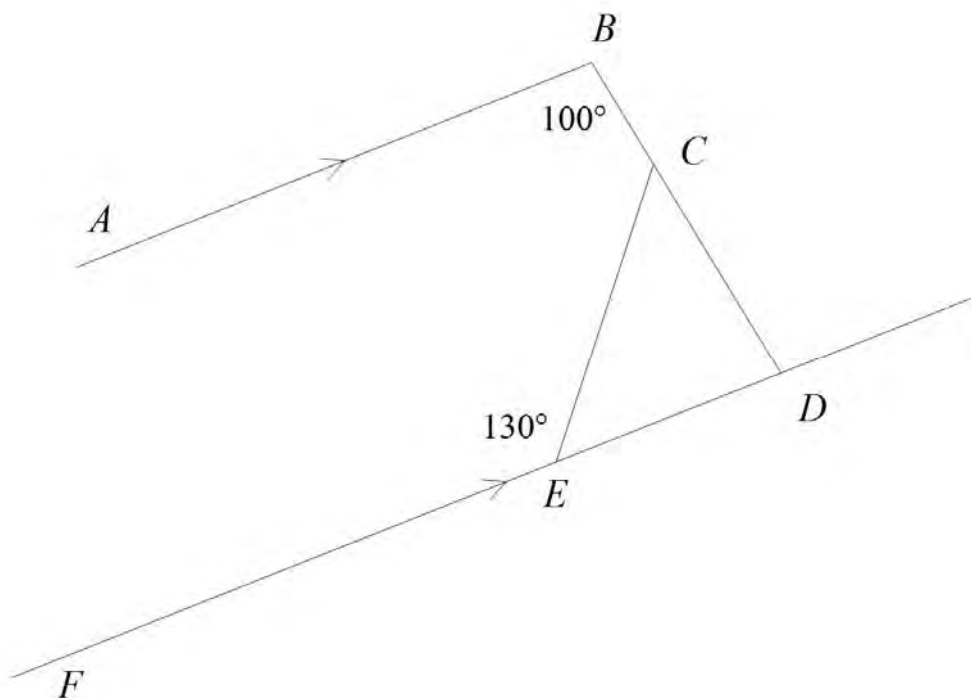


- 3 What is the correct expression for  $AC$  in triangle  $ABC$ ?



- (A)  $\frac{15 \sin 80^\circ}{\sin 40^\circ}$
- (B)  $\frac{15 \sin 80^\circ}{\sin 60^\circ}$
- (C)  $\frac{15 \sin 40^\circ}{\sin 60^\circ}$
- (D)  $\frac{\sin 40^\circ}{15 \sin 80^\circ}$
- 4 The line  $6x - ky = 2$  passes through the point  $(3, 2)$ . What is the value of  $k$ ?
- (A)  $\frac{10}{3}$
- (B)  $-\frac{10}{3}$
- (C)  $-8$
- (D)  $8$

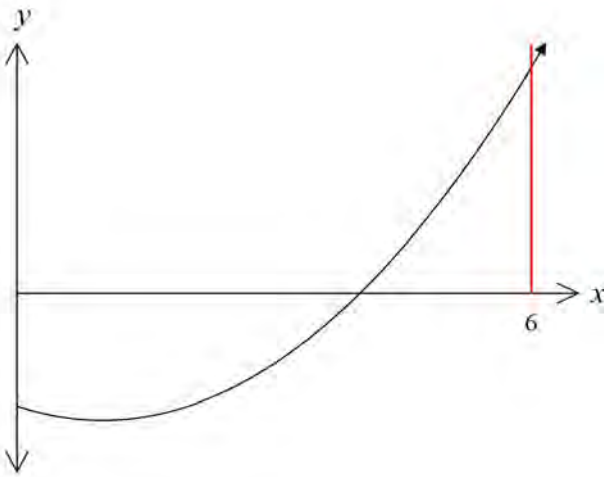
- 5 In the diagram below,  $AB$  is parallel to  $FD$ ,  $\angle ABC = 100^\circ$  and  $\angle CEF = 130^\circ$



What is the value of  $\angle BCE$  ?

- (A)  $100^\circ$
  - (B)  $110^\circ$
  - (C)  $120^\circ$
  - (D)  $130^\circ$
- 6 What is the value of  $f'(3)$  if  $f(x) = 3x - x^3$  ?
- (A)  $f'(3) = -18$
  - (B)  $f'(3) = -24$
  - (C)  $f'(3) = 0$
  - (D)  $f'(3) = 9$

- 7 The diagram below shows part of the graph of  $y = x^2 - 2x - 8$ .



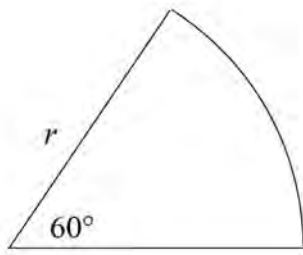
What is the correct expression for the area bounded by the  $x$ -axis and the curve  $y = x^2 - 2x - 8$  between  $0 \leq x \leq 6$ ?

- (A)  $A = \int_0^5 x^2 - 2x - 8 dx + \left| \int_5^6 x^2 - 2x - 8 dx \right|$
- (B)  $A = \int_0^4 x^2 - 2x - 8 dx + \left| \int_4^6 x^2 - 2x - 8 dx \right|$
- (C)  $A = \left| \int_0^5 x^2 - 2x - 8 dx \right| + \int_5^6 x^2 - 2x - 8 dx$
- (D)  $A = \left| \int_0^4 x^2 - 2x - 8 dx \right| + \int_4^6 x^2 - 2x - 8 dx$

- 8 What is the solution to the inequation  $x^2 + 4x + 3 \geq 0$ ?

- (A)  $x \leq 1$  or  $x \geq 3$
- (B)  $x \leq -1$  or  $x \geq -3$
- (C)  $x \geq 1$  or  $x \leq 3$
- (D)  $x \geq -1$  or  $x \leq -3$

- 9 The sector below has an area of  $10\pi$  square units.



Not to scale

What is the value of  $r$ ?

- (A)  $\sqrt{60}$
- (B)  $\sqrt{60\pi}$
- (C)  $\sqrt{\frac{\pi}{3}}$
- (D)  $\sqrt{\frac{1}{3}}$
- 10 An infinite geometric series has a first term of 8 and a limiting sum of 12. What is the common ratio?
- (A)  $\frac{1}{6}$
- (B)  $\frac{1}{4}$
- (C)  $\frac{1}{3}$
- (D)  $\frac{1}{2}$

**Section II****90 marks****Attempt Questions 11-16****Allow about 2 hours and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Question 11-16, your responses should include relevant mathematical reasoning and /or calculations.

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**Question 11** (15 marks) Use the Question 11 Writing Booklet.

- (a) Find the natural domain of the function  $f(x) = \sqrt{3-x}$ . **1**
- (b) Find  $\int x^2 + 1 dx$  **1**
- (c) Solve  $x^2 - x - 1 = 0$  writing your answer in simplest surd form. **2**
- (d) Differentiate  $y = \sqrt{9 - 2x^3}$ . **2**
- (e) Evaluate  $\int_0^3 x^2 - 3 dx$ . **2**
- (f) Evaluate  $\lim_{x \rightarrow 3} \frac{5x - 15}{x^2 + 4x - 21}$ . **2**
- (g) Fully simplify  $(3\sqrt{5} - 2\sqrt{3})^2$ . **2**
- (h) Simplify  $\frac{x-3y}{x^3y} \div \frac{3y-x}{xy^3}$ . **3**

**End of Question 11**

**Question 12** (15 marks) Use the Question 12 Writing Booklet.

(a) If  $f'(x) = 6x^2 + 5x - 1$  and  $f(-1) = 5$ , find an expression for  $f(x)$ . **2**

(b) (i) Show that  $\frac{d}{dx}(x \ln x - x) = \ln x$ . **1**

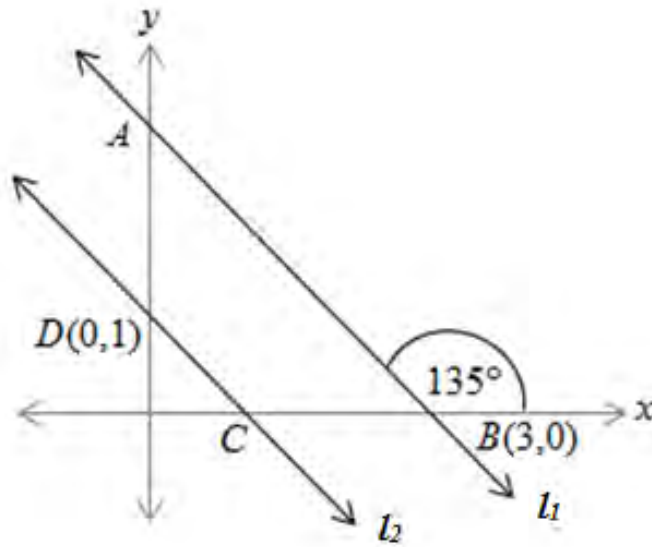
(ii) Hence, or otherwise, evaluate  $\int_1^{e^3} \ln(x) dx$ . Leave your answer in exact form. **2**

(c) A particle moves so that its displacement from the origin is given by  $x = -t^2 + 7t + 8$  (where  $x$  is displacement in metres and  $t$  is time in seconds).

(i) Show that the initial displacement of the particle is 8 metres. **1**

(ii) At what time will the particle be at the origin? **2**

- (d) The line  $l_1$  makes an angle of  $135^\circ$  at the point  $B(3,0)$ . It cuts the  $y$ -axis at  $A$ .  
The line  $l_2$  is parallel to the line  $l_1$ . Its  $y$ -intercept is  $D(0,1)$  and its  $x$ -intercept is  $C$ .



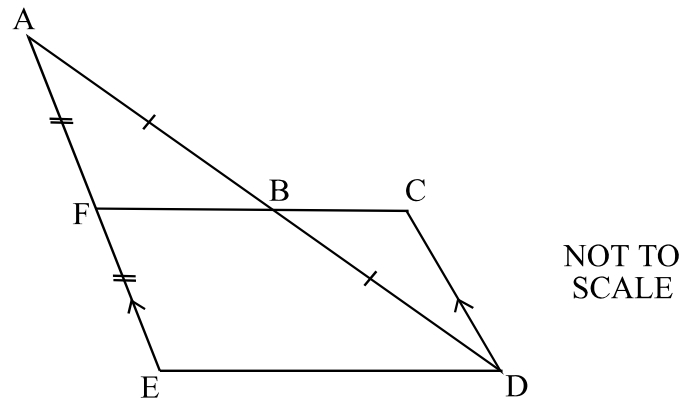
- |                                                                           |   |
|---------------------------------------------------------------------------|---|
| (i) Show that the equation of the line $l_1$ is $x + y - 3 = 0$           | 2 |
| (ii) Hence, or otherwise, find the equation of the line $l_2$ .           | 1 |
| (iii) Find the perpendicular distance between the lines $l_1$ and $l_2$ . | 2 |
| (iv) Find the area of the quadrilateral $ABCD$ .                          | 2 |

**End of Question 12**

**Question 13** (15 marks) Use the Question 13 Writing Booklet.

(a) What is the derivative of  $\frac{e^x}{x^2}$ ? 2

(b) In the diagram, the line  $FC$  bisects  $AE$  at  $F$  and  $AD$  at  $B$ . The line  $AE$  is parallel to  $CD$ .



(i) Explain why  $ED = 2BF$ . 1

(ii) Prove that  $\triangle ABF \equiv \triangle DBC$ . 3

(c) A population of a country grows over time according to  $P = P_0 e^{kt}$ . The population grew from 350 000 in 2001 to 460 000 in 2005. The rate of population growth is proportional to the population size.

(i) Find the growth rate per year, correct to 3 decimal places. 2

(ii) Find the population of the country in 2015, correct to the nearest person. 1

(iii) Calculate the rate of change of the population in 2021. 2

- (d) The area enclosed between the curve  $y = 4 - x^2$  and the line  $y = 4 - 2x$  is rotated about the  $x$ -axis.
- (i) Sketch the region between the two graphs. 2
- (ii) Find the volume of the solid generated between these two graphs, leaving your answer in terms of  $\pi$ . 2

**End of Question 13**

**Question 14** (15 marks) Use the Question 14 Writing Booklet.

- (a) Let  $\alpha$  and  $\beta$  be roots of the equation  $x^2 - 8x + 5 = 0$ . Find the value of  $\alpha^2 + \beta^2$ . **2**
- (b) For the function  $y = xe^{2x}$ ,
- (i) Find the stationary point and determine its nature. **3**
  - (ii) Find any points of inflection. **3**
  - (iii) Sketch the function in the domain  $-3 \leq x \leq 0.5$ . **2**
- (c) Madison is learning to drive. Her first lesson is 10 minutes long. Her second lesson is 15 minutes long. Each subsequent lesson is 5 minutes longer than the previous lesson.
- (i) How long will Madison's fifteenth lesson be? **1**
  - (ii) How many minutes of lessons will Madison have completed after her fifteenth lesson? **2**
  - (iii) During which lesson will Madison have completed a total of 1150 minutes of driving lessons? **2**

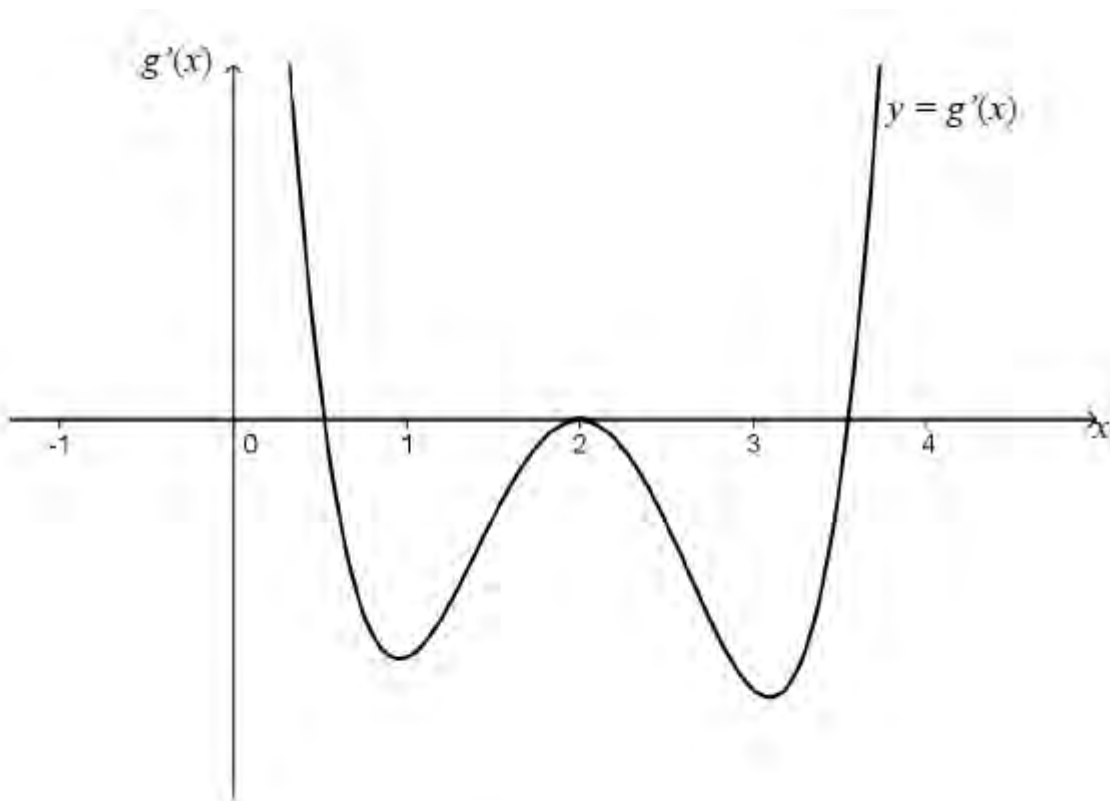
**End of Question 14**

**Question 15** (15 marks) Use the Question 15 Writing Booklet.

(a) What are the exact solutions to the equation  $e^{6x} - 7e^{3x} + 6 = 0$ ? 3

(b) Find the equation of the parabola which has its vertex at  $(2,0)$  and its directrix is given by  $x = 5$ . 2

(c) Shown below is a graph of the derivative function  $y = g'(x)$ .



(i) If the function  $y = g(x)$  were to be drawn using information from the graph above, what feature would exist on the graph at  $x = 2$ ? Justify your answer. 2

(ii) In your answer booklet, draw a neat sketch of a possible function for  $y = g(x)$  given that  $g(0) = 0$ . 2

(d) Alex borrowed \$60 000 to buy a small business. He was charged 6% p.a. compounding monthly on the balance owing and he repaid the loan plus interest in equal monthly repayment over 5 years.

- (i) Show that Alex owed \$  $(60\,000 - M)$  immediately after making his first monthly repayment of \$ $M$ . **1**
- (ii) Show that Alex owed \$  $[ 60\,000(1.005)^3 - M(1.005^2 + 1.005 + 1) ]$  immediately after he made three monthly repayments. **2**
- (iii) Calculate his monthly repayment, \$ $M$  to the nearest five cents. **2**
- (iv) Calculate the total amount of interest paid. **1**

**End of Question 15**

**Question 16** (15 marks) Use the Question 16 Writing Booklet.

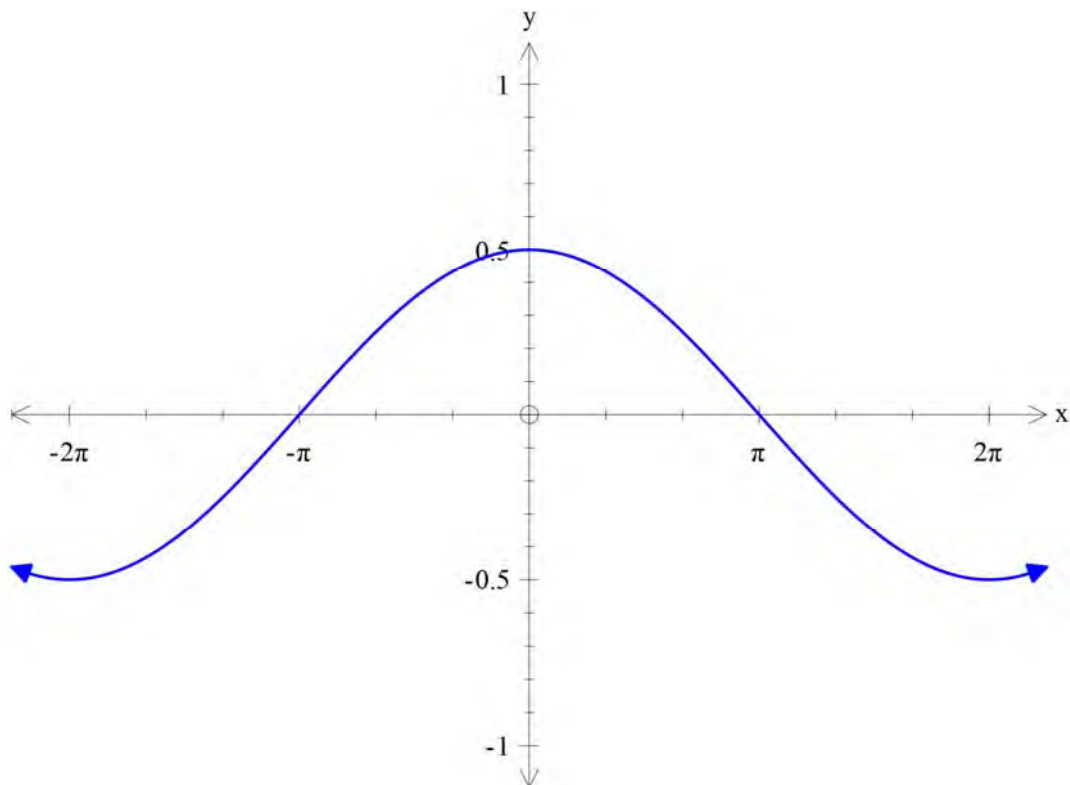
(a) For the curve  $y = \ln(x - 2)$ ,

(i) Sketch the curve. 1

(ii) State its domain and range. 2

(b) Sketch the function  $y = -3 \sin \frac{x}{2}$  in the domain  $-2\pi \leq x \leq 4\pi$  2

(c) For the function  $y = \frac{1}{2} \cos \frac{x}{2}$  below,



(i) Find the area between the curve and the lines  $x = \frac{\pi}{2}$  and  $x = 2\pi$ . 3

(ii) Use one application of Simpson's Rule to find an approximation for the area between the curve and the lines  $x = -\pi$  and  $x = \pi$ . 2

(iii) What is the percentage error of your approximation from (ii) compared with the actual area between the curve and the lines  $x = -\pi$  and  $x = \pi$ ? 1

- (d) The area enclosed by the curve  $y = \sqrt{r^2 - x^2}$  is rotated about the  $x$ -axis.
- (i) What is the name given to the solid that is generated? **1**
- (ii) Explain why the volume of the solid of revolution between  $x = -r$  and  $x = r$  is **twice** the integral  $\pi \int_0^r (r^2 - x^2) dx$ . **1**
- (iii) Show that the volume of the solid formed is  $\frac{4}{3} \pi r^3$  **2**

**End of paper**

**ANSWER SHEET FOR MULTIPLE CHOICE SECTION****Student Exam number:** \_\_\_\_\_**Teacher:** \_\_\_\_\_

1.    A ○ B ○ C ○ D ○
2.    A ○ B ○ C ○ D ○
3.    A ○ B ○ C ○ D ○
4.    A ○ B ○ C ○ D ○
5.    A ○ B ○ C ○ D ○
6.    A ○ B ○ C ○ D ○
7.    A ○ B ○ C ○ D ○
8.    A ○ B ○ C ○ D ○
9.    A ○ B ○ C ○ D ○
10.   A ○ B ○ C ○ D ○



**STANDARD INTEGRALS**

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

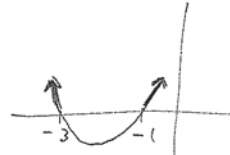
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | comments |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>Multiple Choice</u></p> <p>1) ans = 0.04099...<br/>= 0.0410 (3sf) → (B)</p> <p>2) (D)</p> <p>3) <math>\frac{AC}{\sin 80} = \frac{15}{\sin 40}</math><br/><math>AC = \frac{15 \sin 80}{\sin 40} \rightarrow (A)</math></p> <p>4) <math>6(3) - (2)K = 2</math><br/><math>-2K = -16</math><br/><math>K = 8 \rightarrow (D)</math></p> <p>5) (D)</p> <p>6) <math>f'(x) = 3 - 3x^2</math><br/><math>f'(3) = 3 - 3(3)^2</math><br/><math>= 3 - 27</math><br/><math>= -24 \rightarrow (B)</math></p> <p>7) <math>x^2 - 2x - 8 = 0</math><br/><math>(x-4)(x+2) = 0</math><br/><math>x = (4), -2 \rightarrow (D)</math></p> |          |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | comments |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p>8) <math>x^2 + 4x + 3 \geq 0</math><br/><math>(x+1)(x+3) \geq 0</math></p>  <p><math>x \leq -3, x \geq -1 \rightarrow (D)</math></p> <p>9) <math>A = \frac{1}{2} r^2 \theta</math><br/><math>10\pi = \frac{1}{2} r^2 \frac{\pi}{3}</math><br/><math>\frac{r^2}{2} = 30</math><br/><math>= 60</math><br/><math>r = \sqrt{60} \rightarrow (A)</math></p> <p>10) <math>\sum_{n=0}^{\infty} r^n = \frac{a}{1-r}</math><br/><math>12 = \frac{8}{1-r}</math><br/><math>12 - 12r = 8</math><br/><math>-12r = -4</math><br/><math>r = \frac{1}{3} \rightarrow (C)</math></p> |          |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | comments                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Question 11</p> <p>a) <math>f(x) = \sqrt{3-x}</math></p> <p><math>3-x \geq 0</math><br/> <math>-x \geq -3</math><br/> <math>x \leq 3</math></p> <p>b) <math>\int x^2 + 1 \, dx</math></p> <p><math>= \frac{x^3}{3} + x + C</math></p> <p>c) <math>x^2 - x - 1 = 0</math></p> <p><math>x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}</math></p> <p><math>= \frac{1 \pm \sqrt{1 - 4(1)(-1)}}{2}</math></p> <p><math>= \frac{1 \pm \sqrt{5}}{2}</math></p> <p>d) <math>y = \sqrt{9-2x^3} = (9-2x^3)^{\frac{1}{2}}</math></p> <p><math>y' = \frac{1}{2}(9-2x^3)^{-\frac{1}{2}} \cdot -6x^2</math></p> <p><math>= -3x^2(9-2x^3)^{-\frac{1}{2}}</math></p> <p>OR <math>= -\frac{3x^2}{\sqrt{9-2x^3}}</math></p> | <p>Some students failed to realise you cannot take the square root of a negative. i.e. <math>3-x \geq 0</math>.</p> <p>good.</p> <p>good.</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | comments                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <p>e) <math>\int_0^3 x^2 - 3 \, dx</math></p> <p><math>= \left[ \frac{x^3}{3} - 3x \right]_0^3</math></p> <p><math>= \frac{(3)^3}{3} - 3(3) - 0</math></p> <p><math>= 9 - 9</math></p> <p><math>= 0</math></p> <p>f) <math>\lim_{x \rightarrow 3} \frac{5x-15}{x^2+4x-21}</math></p> <p><math>= \lim_{x \rightarrow 3} \frac{5(x-3)}{(x+7)(x-3)}</math></p> <p><math>= \frac{5}{3+7}</math></p> <p><math>= \frac{1}{2}</math></p> <p>g) <math>(3\sqrt{5} - 2\sqrt{3})^2</math></p> <p><math>= (3\sqrt{5})^2 + 2 \times (3\sqrt{5})(-2\sqrt{3}) + (-2\sqrt{3})^2</math></p> <p><math>= 45 - 12\sqrt{15} + 12</math></p> <p><math>= 57 - 12\sqrt{15}</math></p> | <p>good.</p> <p>good.</p> <p>many failed to first factorise then sub in <math>x=3</math>.</p> <p>good.</p> |

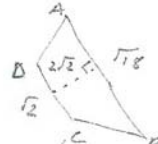
Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                             | comments                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <p>h) <math>\frac{x-3y}{x^3y} \div \frac{3y-x}{xy^3}</math></p> <p><math>= \frac{x-3y}{x^3y} \times \frac{xy^3}{3y-x}</math></p> <p><math>= \frac{x-3y}{x^2} \times \frac{y^2}{-(x-3y)}</math></p> <p><math>= -\frac{y^2}{x^2}</math></p>                                                         | <p>some failed to take out -1 here in order to cancel.</p> |
| Question 12                                                                                                                                                                                                                                                                                       |                                                            |
| <p>a) <math>f'(x) = 6x^2 + 5x - 1</math></p> <p><math>f(x) = \frac{26x^3}{3} + \frac{5x^2}{2} - x + C</math></p> <p><math>5 = 2(-1)^3 + \frac{5(-1)^2}{2} - (-1) + C</math></p> <p><math>C = \frac{7}{2}</math></p> <p><math>\therefore f(x) = 2x^3 + \frac{5x^2}{2} - x + \frac{7}{2}</math></p> | <p>answered ok by most.</p>                                |
| <p>b) i) <math>\frac{d}{dx}(x \ln x - x) = u'v + v'u - 1</math></p> <p><math>= \ln x + \frac{1}{x} \times x - 1</math></p> <p><math>= \ln x + 1 - 1</math></p> <p><math>= \ln x</math> as req'd.</p>                                                                                              | <p>many failed to use product rule.</p>                    |
| <p>ii) <math>\int_1^{e^3} \ln(x) dx = [x \ln x - x]_1^{e^3}</math></p> <p><math>= (e^3 \ln e^3 - e^3) - (\ln 1 - 1)</math></p> <p><math>= 3e^3 - e^3 - -1</math></p> <p><math>= 2e^3 + 1</math></p>                                                                                               | <p>students could not relate part (i) to part (ii)</p>     |

Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | comments                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>c) <del>are</del></p> <p>i) initial displacement is x when <math>t=0</math></p> <p><math>\therefore x = -(0)^2 + 7(0) + 8</math></p> <p><math>= 8m</math></p> <p>ii) i.e. find t when <math>x=0</math></p> <p><math>-t^2 + 7t + 8 = 0</math></p> <p><math>-(t^2 - 7t - 8) = 0</math></p> <p><math>(t-8)(t+1) = 0</math></p> <p><math>t = 8, -1</math> (<math>t \geq 0</math>)</p> <p><math>\therefore</math> at the origin when <math>t = 8</math> seconds.</p>                                     | <p>well answered.</p>                                                                                                                                     |
| <p>d) i) <math>m_1 = \tan \theta</math></p> <p><math>= \tan 135^\circ</math></p> <p><math>= -1</math></p> <p><math>l_1: y - y_1 = m(x - x_1)</math></p> <p><math>y - 0 = -1(x - 3)</math></p> <p><math>y = -x + 3</math></p> <p><math>x + y - 3 = 0</math></p> <p>ii) <math>m_{l_1} = m_{l_2} = -1</math></p> <p><math>l_2: y - y_1 = m(x - x_1)</math></p> <p><math>y - 1 = -1(x - 0)</math></p> <p><math>y - 1 = -x</math></p> <p><math>x + y - 1 = 0</math></p> <p>(or <math>y = -x + 1</math>)</p> | <p>many failed to state <math>t \geq 0</math>.</p> <p>co-ord geom very poorly answered, students failed to recall formulae + relate grad of 11 lines.</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | comments                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <p>iii) <math>d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}</math> (using D(0,1) and <math>d_1</math>)</p> $= \frac{ 1 - 3 }{\sqrt{1^2 + 1^2}}$ $= \frac{2}{\sqrt{2}}$ $= \sqrt{2} \text{ units}$ <p>iv) <math>A = \frac{1}{2}h(a+b)</math></p> $d_{DL} = \sqrt{2}$ $d_{AB} = \sqrt{18} = 3\sqrt{2}$ $\therefore A = \frac{1}{2} \times \sqrt{2} (\sqrt{2} + 3\sqrt{2})$ $= \sqrt{2} (\sqrt{2} + 3\sqrt{2}) \div 2$ $= \sqrt{2} \times 4\sqrt{2} \div 2$ $= 8 \div 2$ $= 4 \text{ units}^2$  | <p>poor use of correct formula</p> <p>Answered.<br/>OK<br/>with exception of a few who failed to use correct formula</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

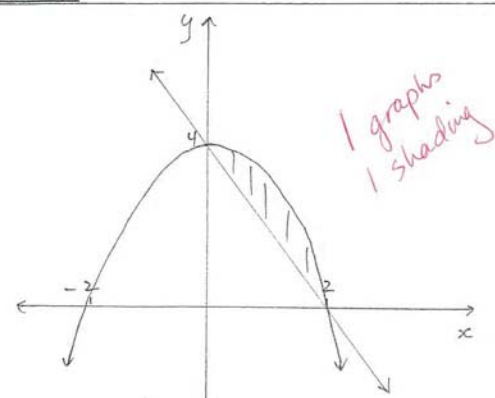
| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | comments                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Question 13</p> <p>a) <math>\frac{d}{dx} \left( \frac{e^x}{x^2} \right) = \frac{u'v - v'u}{v^2}</math></p> $= \frac{e^x x^2 - 2x e^x}{x^4}$ $= \frac{e^x x (x - 2)}{x^4}$ $= \frac{e^x (x - 2)}{x^3}$ <p>b) i) <math>\triangle AFB \parallel \triangle AED</math> (equilateral)</p> $\frac{AF}{AE} = \frac{AB}{AD} = \frac{1}{2}$ $\therefore ED = 2BF$ <p>ii) <math>A: \angle ABF = \angle CBD</math> (vert. opp. <math>\angle</math>'s)</p> $A: \angle AFB = \angle BCD$ (alt. $\angle$ 's equal) $S: AB = BD$ (given) $\therefore \triangle ABF \equiv \triangle BDC$ (AAS) <p>(or SAS, <math>FB = BC</math>)</p> | <p>Several students wrote this as a product, but were unsuccessful in the derivative of a product <math>\Rightarrow</math> learn the rules.</p> <p>Other versions accepted.</p> <p>And other versions</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                               | comments                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <p>c) i) <math>P = P_0 e^{kt}</math><br/> <math>P_0 = 350,000</math>. when <math>t=4</math>, <math>P = 460,000</math></p> $460,000 = 350,000 e^{4k}$ $e^{4k} = \frac{46}{35}$ $\log_e e^{4k} = \log_e \left( \frac{46}{35} \right)$ $4k = \ln \frac{46}{35}$ $k = \ln \frac{46}{35} \div 4$ $= 0.068 \text{ (3dp)}$ | <p>Usually well done, although some students used <math>t=5</math>.</p> |
| <p>ii) i.e. find <math>P</math> when <math>t=14</math></p> $P = 350,000 e^{14k}$ $= 910,924.32 \dots$ $= 910,924 \text{ (nearest person)}$                                                                                                                                                                          | <p>ECF</p>                                                              |
| <p>iii) i.e. find <math>\frac{dP}{dt}</math> when <math>t=20</math></p> $\frac{dP}{dt} = kP$ $= k P_0 e^{kt}$ $= k 350,000 e^{20k}$ $= 93,774 \text{ people / yr}$                                                                                                                                                  | <p>ECF.</p>                                                             |

accept 92729 people for using  $k=0.068$ .

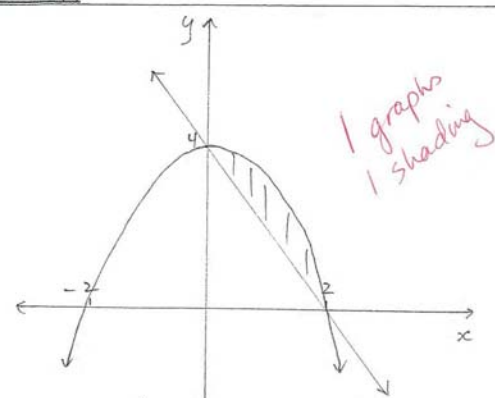
## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | comments                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <p>d) i)</p>  <p>ii)</p> $V = \pi \int_a^b y^2 dx$ $= \pi \int_{-2}^2 (4-x^2)^2 dx = \pi \int_{-2}^2 (4-2x)^2 dx$ $= \pi \int_{-2}^2 16 - 8x^2 + x^4 - (16 - 16x + 4x^2) dx$ $= \pi \int_{-2}^2 16 - 8x^2 + x^4 - 16 + 16x - 4x^2 dx$ $= \pi \int_{-2}^2 x^4 - 12x^2 + 16x dx$ $= \pi \left[ \frac{x^5}{5} - \frac{12x^3}{3} + \frac{16x^2}{2} \right]_0^2$ $= \pi \left( \frac{(2)^5}{5} - 4(2)^3 + 8(2)^2 \right)$ $= \frac{32\pi}{5} \text{ units}^3$ | <p>Expansions were badly done and compounded errors.</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s) | Question 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | comments                                                                                 |       |      |      |    |       |   |       |                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------|------|------|----|-------|---|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a)                    | $x^2 - 8x + 5 = 0$<br>$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$<br>$= \left(-\frac{6}{1}\right)^2 - 2\left(\frac{5}{1}\right)$<br>$= \left(-\frac{9}{1}\right)^2 - 2\left(\frac{5}{1}\right)$<br>$= 64 - 10$<br>$= 54$                                                                                                                                                                                                                                                                                                                     | Mostly well done, but students must be able to find $\alpha + \beta$ and $\alpha\beta$ . |       |      |      |    |       |   |       |                                                                                                                                                                 |
| b)                    | $y = xe^{2x}$<br>i) $y' = u'v + v'u$<br>$= e^{2x} + 2e^{2x}x$<br>$= e^{2x}(1 + 2x)$<br>SP's exist when $y' = 0$ .<br>$e^{2x}(1 + 2x) = 0$<br>$e^{2x} = 0$ or $1 + 2x = 0$<br>$\downarrow$<br>(no soln.) $x = -\frac{1}{2}$<br><table border="1"><tr><td>x</td><td>-0.6</td><td>-0.5</td><td>-0.4</td></tr><tr><td>y'</td><td>-0.06</td><td>0</td><td>+0.09</td></tr></table><br>when $x = -\frac{1}{2}$ , $y = -\frac{1}{2}e^{2x - \frac{1}{2}}$<br>$= -\frac{1}{2e}$<br>$\therefore$ minimum turning point at $\left(-\frac{1}{2}, -\frac{1}{2e}\right)$ | x                                                                                        | -0.6  | -0.5 | -0.4 | y' | -0.06 | 0 | +0.09 | This part was reasonably well done<br><br>Errors here with the negative being omitted<br><br>Some students did not find the y value or determine max/min value. |
| x                     | -0.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -0.5                                                                                     | -0.4  |      |      |    |       |   |       |                                                                                                                                                                 |
| y'                    | -0.06                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                                                                                        | +0.09 |      |      |    |       |   |       |                                                                                                                                                                 |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                              | comments                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| d) i)                                                                                                                                                                                                                                                                                                                                                                           |                                                          |
| ii) $V = \pi \int_{-2}^2 y^2 dx$<br>$= \pi \int_{-2}^2 (4 - x^2)^2 dx - \pi \int_{-2}^2 (4 - 2x)^2 dx$<br>$= \pi \int_{-2}^2 16 - 8x^2 + x^4 - (16 - 16x + 4x^2) dx$<br>$= \pi \int_{-2}^2 16 - 8x^2 + x^4 - 16 + 16x - 4x^2 dx$<br>$= \pi \int_{-2}^2 x^4 - 12x^2 + 16x dx$<br>$= \pi \left[ \frac{x^5}{5} - \frac{12x^3}{3} + \frac{16x^2}{2} \right]_{-2}^2$<br>$= \pi \left( \frac{(2)^5}{5} - 4(2)^3 + 8(2)^2 \right)$<br>$= \frac{32\pi}{5} \text{ units}^3$ | <p>Expansions were badly done and compounded errors.</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | comments |      |      |      |       |       |   |      |                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|------|------|-------|-------|---|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>ii) <math>y' = e^{2x}(1+2x)</math><br/><math>y'' = u'v + v'u</math><br/><math>= 2e^{2x}(1+2x) + 2e^{2x}</math><br/><math>= 2e^{2x}(1+2x+1)</math><br/><math>= 2e^{2x}(2+2x)</math><br/><math>= 4e^{2x}(1+x)</math></p> <p>POI's exist when <math>y''=0</math> and concavity changes.</p> <p><math>4e^{2x}(1+x) = 0</math><br/><math>4e^{2x} = 0</math> or <math>1+x = 0</math><br/><math>\downarrow</math> no soln. <math>x = -1</math></p> <table border="1"><tr><td><math>x</math></td><td>-1.1</td><td>-1</td><td>-0.9</td></tr><tr><td><math>y''</math></td><td>-0.04</td><td>0</td><td>0.07</td></tr></table> <p>when <math>x = -1</math>, <math>y = -e^{-2} = -\frac{1}{e^2}</math><br/><math>\therefore</math> POI exists at <math>(-1, -\frac{1}{e^2})</math></p> <p>iii) when: <math>x = -3</math>, <math>y = -0.007</math><br/><math>x = 0.5</math>, <math>y = 1.36</math></p> <p>1 endpoint<br/>1 other detail.</p> | $x$      | -1.1 | -1   | -0.9 | $y''$ | -0.04 | 0 | 0.07 | <p>some very bad errors here.</p> <p>change in concavity went unchecked.</p> <p>Minus sign left out.</p> <p>Watch transcription of -ve</p> <p>Not well answered, generally</p> |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -1.1     | -1   | -0.9 |      |       |       |   |      |                                                                                                                                                                                |
| $y''$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -0.04    | 0    | 0.07 |      |       |       |   |      |                                                                                                                                                                                |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | comments                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>c) 10, 15, 20, ...</p> <p>i) AP with <math>a=10</math>, <math>d=5</math><br/> <math>T_{15} = a + (n-1)d</math><br/> <math>= 10 + 5(15-1)</math><br/> <math>= 80 \text{ mins.}</math></p> <p>ii) i.e. find <math>S_{15}</math><br/> <math>S_n = \frac{n}{2}(a+d)</math><br/> <math>S_{15} = \frac{15}{2}(10+80)</math><br/> <math>= 675 \text{ min.}</math><br/> <math>\rightarrow 11 \text{ h } 15 \text{ min.}</math></p> <p>iii) i.e. find <math>n</math> when <math>S_n = 1150</math><br/> <math>S_n = \frac{n}{2}(2a + (n-1)d)</math><br/> <math>1150 = \frac{n}{2}(20 + 5(n-1))</math><br/> <math>= \frac{n}{2}(20 + 5n - 5)</math><br/> <math>= \frac{n}{2}(15 + 5n)</math><br/> <math>2300 = 15n + 5n^2</math><br/> <math>5n^2 + 15n - 2300 = 0</math><br/> <math>n^2 + 3n - 460 = 0</math><br/> <math>(n+23)(n-20) = 0</math><br/> <math>\therefore n = 20</math> (i.e. the 20th lesson)</p> | <p>Good</p> <p>Good.</p> <p>Error in question:<br/>1150 is in minutes, not hours.</p> <p>Marking was awarded for this version of working and also for students working on merit (i.e. correct units worked through as best as possible).</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

## Suggested solution(s)

Question 15

a)  $e^{3x} - 7e^{3x} + 6 = 0$

let  $u = e^{3x}$

$u^2 - 7u + 6 = 0$

$(u-6)(u-1) = 0$

$u = 1, 6$

(3)

$\therefore e^{3x} = 1, \quad e^{3x} = 6$

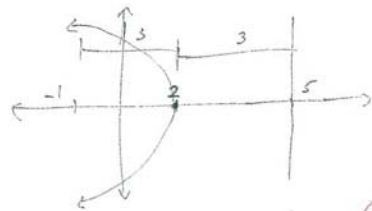
$\ln e^{3x} = \ln 1, \quad \ln e^{3x} = \ln 6$

$3x = \ln 1, \quad 3x = \ln 6$

$x = \frac{\ln 1}{3}, \quad x = \frac{\ln 6}{3}$

$= 0$

b)



$V = (2, 0)$

$a = 3$

(2)

$\therefore (y-0)^2 = -4 \times 3 (x-2)$

$y^2 = -12(x-2)$

Many students  
got to  $1+6$   
but couldn't  
go further.

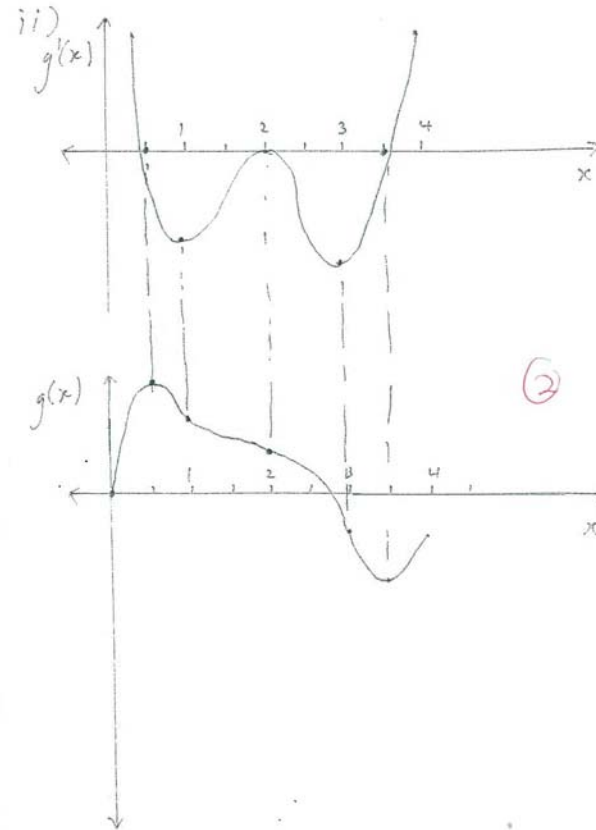
few mixed  
 $x^2 = 4a(y-b)!$

## Suggested Solutions, Marking Scheme and Markers' comments

## Suggested solution(s)

$$\begin{array}{c} \text{c) i)} \end{array} \begin{array}{|c|c|c|c|} \hline y & 1.9 & 2 & 2.1 \\ \hline g'(x) & - & 0 & - \\ \hline \end{array}$$

$\therefore$  horizontal point of inflection  
at  $x=2$  since  $g'(2) = 0$  and (2)  
 $g'(x) < 0$  on either side of  $x=2$ .



(2)

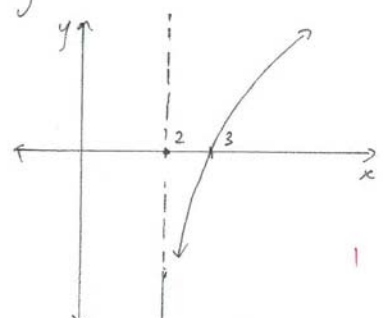
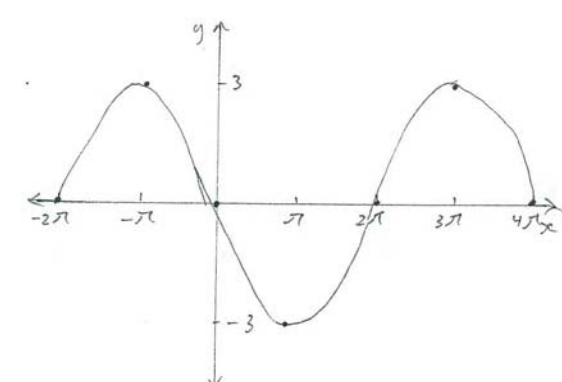
many stated  
SP, no follow  
through

very poorly  
answered as  
student failed  
to interpret  
 $g'(x)$  to  $f(x)$   
esp point  $x=$

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | comments                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <p>d) <math>r = 0.06 \div 12 = 0.005</math></p> <p>i) <math>A_1 = P(1+r)^1 - M</math><br/> <math>= 60000(1.005)^1 - M</math><br/> <math>= \\$ (60300 - M)</math> 1</p> <p>ii) <math>A_3 = (P+I) - (R+I)</math><br/> <math>P+I = 60000(1.005)^3</math> 2<br/> <math>R+I = M + M(1.005)^1 + M(1.005)^2</math><br/> <math>= M(1.005^2 + 1.005 + 1)</math><br/> <math>\therefore A_3 = \\$ \left( 60000(1.005)^3 - M(1.005^2 + 1.005 + 1) \right)</math></p> <p>iii) <math>A_{60} = 0</math><br/> <math>0 = 60000(1.005)^{60} - \frac{M(1.005^{60} - 1)}{0.005}</math><br/> <math>M(1.005^{60} - 1) = 60000(1.005)^{60} \times 0.005</math> 2<br/> <math>M = \\$1159.95</math></p> <p>iv) Interest = <math>(1159.95 \times 60) - 60000</math><br/> <math>= \\$7597.</math> 1</p> | <p>✓</p> <p>✓</p> <p>most ans<br/>v. well.</p> <p>poorly answered.</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | comments                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Question 16</p> <p>a) <math>y = \ln(x-2)</math></p> <p>i)  1</p> <p>ii) domain: <math>x &gt; 2</math> 1<br/> range: <math>\text{all real } y</math> 1</p> <p>b) <math>y = -3 \sin \frac{x}{2}, -2\pi \leq x \leq 4\pi</math><br/> amp = -3, period = <math>\frac{2\pi}{\frac{1}{2}} = \frac{2\pi}{1/2} = 4\pi</math></p> <p> 2</p> | <p>Some failed to show the asymptote at <math>x=2</math>.</p> <p>✓ done well. 'all' marks awarded</p> <p>Some used +3 as the amplitude or failed to realise period was <math>4\pi</math>.</p> |

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | comments                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| c) $y = \frac{1}{2} \cos \frac{x}{2}$<br>ii) $A = \int_{\pi/2}^{\pi} \frac{1}{2} \cos \frac{x}{2} dx + \left  \int_{\pi}^{2\pi} \frac{1}{2} \cos \frac{x}{2} dx \right $<br>$= \left[ \sin \frac{x}{2} \right]_{\pi/2}^{\pi} + \left  \left[ \sin \frac{x}{2} \right]_{\pi}^{2\pi} \right $<br>$= \left( \sin \frac{\pi}{2} - \sin \frac{\pi}{4} \right) + \left  \left( \sin \pi - \sin \frac{\pi}{2} \right) \right $<br>$= \left( 1 - \frac{1}{\sqrt{2}} \right) + \left  (0 - 1) \right $<br>$= 1 - \frac{1}{\sqrt{2}} + 1$<br>$= 2 - \frac{1}{\sqrt{2}}$<br>$= \frac{4 - \sqrt{2}}{2} \text{ units}^2$ | - Some calculated the integral: $\int_{2\pi}^{\pi}$ for which 2 $\pi/2$ each marks was awarded if evaluated correctly. |
| OR<br>ii) $A \div \frac{\pi}{3} \left( \frac{1}{2} \cos -\frac{\pi}{2} + 4 \times \frac{1}{2} \cos \frac{0}{2} + \frac{1}{2} \cos \frac{\pi}{2} \right)$<br>$\div \frac{\pi}{3} (0 + 2 + 0)$<br>$\div \frac{2\pi}{3} \text{ units}^2$                                                                                                                                                                                                                                                                                                                                                                       | - Some failed to use $\frac{2}{3}\pi$ as the width of the strip.                                                       |
| iii) $\frac{\frac{2\pi}{3} - 2}{2} \times 100 = 4.7\% (1 \text{ dp})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | - each mark awarded if above area used.                                                                                |

i) 1.29 ii) 2.09

## Suggested Solutions, Marking Scheme and Markers' comments

| Suggested solution(s)                                                                                                                                                                                                                                                                                                                 | comments                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| d) i) Sphere<br>ii) $y = \sqrt{r^2 - x^2}$ is even / symmetrical about y-axis<br>$\int$ produces hemi-sphere<br>$\int$ produces a sphere.                                                                                                                                                                                             | Some said circle.<br>Needed to state the curve is <u>even</u> .     |
| iii) $\pi \int_{-r}^r (r^2 - x^2) dx$<br>$= \pi \left[ r^2 x - \frac{x^3}{3} \right]_{-r}^r$<br>$= \pi \left( r^3 - \frac{r^3}{3} - \left( -r^3 - \frac{-r^3}{3} \right) \right)$<br>$= \pi \left( \frac{2}{3} r^3 - -\frac{2}{3} r^3 \right)$<br>$= \pi \left( \frac{4}{3} r^3 \right)$<br>$= \frac{4}{3} \pi r^3 \text{ as req'd.}$ | The variable here is x not r.<br>i.e. $\int r^2 \neq \frac{r^3}{3}$ |